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REPORT OVERVIEW

This report presents the practical experience and knowledge gained during my Students Industrial Work Experience Scheme (SIWES) at African Steel Mills Nigeria Limited, Ikorodu, Lagos, in the Information Technology Department. The training provided practical exposure to the application of Information Technology in an industrial environment and helped bridge the gap between academic knowledge and workplace practice.

The report is divided into five chapters. Chapter One introduces the Students Industrial Work Experience Scheme, its objectives, and the relevance of industrial training to Information Technology students. Chapter Two presents an overview of African Steel Mills Nigeria Limited, including its organizational structure, departments, operations, and the Information Technology Department where the training was undertaken.

Chapter Three describes the work carried out during the attachment and the experience gained. It covers activities such as computer hardware and software maintenance, networking, CCTV systems, printer and plotter maintenance, biometric and access-control systems, system administration, virtualization, Linux, server and storage infrastructure, and general IT support.

Chapter Four presents the equipment, tools, software and technologies encountered during the training. It describes their functions and how they were used during the attachment, including network switches, routers, SFPs, patch cords, crimping tools, LAN testers, CCTV equipment, computer hardware, printers, plotters, Avaya equipment, Cisco Packet Tracer, VirtualBox, Active Directory, Ubuntu and other IT tools.

Chapter Five summarizes the attachment activities, discusses problems encountered during the programme, presents suggestions for improving the SIWES scheme, and concludes the report by highlighting the practical knowledge and experience gained during the training.

ACKNOWLEDGEMENT

I sincerely acknowledge the Lord Almighty for His guidance, protection, and provision throughout the period of my Student Industrial Work Experience Scheme (SIWES).

I express my gratitude to the management of African Steel Mills Nigeria Limited, Ikorodu, Lagos, for granting me the opportunity to undergo my training in the Information Technology department. I especially thank my supervisors for their patience, support, and willingness to share their knowledge and expertise, which greatly enhanced my learning experience.

I also wish to appreciate my institution, University of Ilorin, and the Industrial Training Fund (ITF) for coordinating the SIWES program and ensuring its smooth implementation. I especially thank the department of Information Technology. Their academic guidance and the department's curriculum provided the theoretical knowledge that enabled me to carry out my industrial assignments with analytical clarity.

Finally, I extend my heartfelt gratitude to my parents, family and friends for their encouragement and support in finances, physically and emotionally through this journey.

ABSTRACT

This report presents the experience and knowledge gained during the Students Industrial Work Experience Scheme (SIWES) undertaken as part of the requirements for the study of Information Technology. The training provided practical exposure to various Information Technology operations within an industrial environment, with emphasis on computer hardware and software maintenance, networking, CCTV systems, printer and plotter maintenance, biometric and access-control systems, telephony, server and storage technologies, and technical user support.

During the training period, practical activities included preventive maintenance of computer systems, hardware installation and troubleshooting, network connectivity testing, Ethernet cable termination and testing, switch and fibre-optic infrastructure maintenance, CCTV installation and troubleshooting, printer and plotter maintenance, and support for biometric systems. The training also provided practical experience with technologies and tools such as Active Directory, virtual machines, Ubuntu Linux, Cisco Packet Tracer, Synology, Navision, Geniac, VNC, Avaya communication systems, and Windows system maintenance utilities. Networking exercises involving subnetting, DHCP, VLANs, routing, and network troubleshooting further strengthened the understanding of computer networks and system administration.

The SIWES programme provided an opportunity to apply theoretical knowledge acquired in the classroom to real-world technical problems. It also developed practical skills in troubleshooting, preventive maintenance, equipment installation, network administration, system configuration, documentation, and technical support. Overall, the experience improved the understanding of how Information Technology infrastructure is deployed, maintained, and supported within an industrial organisation and contributed significantly to the development of practical and professional skills.

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CHAPTER ONE: INTRODUCTION

1.1 Background

The Student Industrial Work Experience Scheme (SIWES) is a skill training programme established by the Industrial Training Fund (ITF) in 1973 to bridge the gap between the theoretical knowledge acquired in the academic institutions and practical skills acquired academic institutions in tertiary institutions, particularly those in science, engineering and technology-related disciplines, including Information Technology.

As a student of the Department of Information Technology, University of Ilorin. I was mandated to undergo six (6) months of industrial attachment as part of the requirements for the award of my degree. This attachment was undertaken at African Steel Mills Nigeria Limited, where I was exposed to practical, hands-on experience in various aspects of information technology operations within an industrial manufacturing environment.

During the period of attachment, I was involved in network support duties, CCTV system installation and support, basic system administration tasks, hardware troubleshooting, as well as help desk support services. This experience enabled me to apply the theoretical concepts learnt in school to real-life industrial IT challenges, while also acquiring new skills that are not typically covered in the classroom.

1.2 Objectives

The general objectives of SIWES are:

- To provide students with practical industrial skills relevant to their course of study.
- To expose students to work methods, techniques, and equipment not available in their institutions.
- To prepare students for the work environment they will encounter after graduation.
- To bridge the gap between classroom theory and industrial practice.
- To facilitate the transition from school to gainful employment.
- To strengthen collaboration between industries and educational institutions.
- To enhance students, employability and professional competence.

CHAPTER TWO: DESCRIPTION OF THE ESTABLISHMENT OF ATTACHMENT

2.1 Location and the brief history of the establishment.

African Steel Mill Nigeria Limited is a steel manufacturing company located at plot 337, Ikorodu Industrial Estate, Ikorodu, Lagos State, Nigeria. The company was incorporated as a Limited Liability company on the 7th of November, 2000, and has grown to become one of the pioneers in Nigeria dealing in the trading and manufacturing of steel products.

African Steel Mills is a wholly-owned subsidiary of African Steel Holdings, which itself is part of the larger African Industries Group (AIG), a multinational conglomerate with diverse business interests spanning steel, glass, plastics, lead, plywood, aluminium products, chemical, and power generation. The parent group, which commenced operations in Nigeria in 1971, has grown to become the country's largest producer, controlling a significant share of about 60% of Nigeria's steel industry. African industrials currently has twelve subsidiaries including six major steel plants such as African Steel Mills, African Industries, Ikorodu Steel Mills, and Abuja Steel Mills, which together produce about one million tonnes of steel annually.

As its Ikorodu facility, African Steel Mills operates a fully integrated steel manufacturing plant with combined melting and rolling production capacity of 225,000 metric tonnes per annum. The company's operation covers the production, rolling, and forging of various types of manufacturing steel and steel products, positioning it as one of the leading steel manufacturers.

Over the years, the company has continued to invest in modernizing its production facilities and expanding its operational capacity. The African Industries Group as a whole as invested over \$1.1 billion into the Nigerian economy, part of which as gone into strengthening subsidiaries like African Steel Mills through equipment upgrades and backward integration initiatives aimed at reducing dependence on imported raw materials.

Today, African Steel Mills Nigeria Limited stands as a key player in Nigeria's industrial and manufacturing sector, contributing significantly to the local availability of steel products used in construction, infrastructure development, and various downstream manufacturing activities across the country.

2.2 Objectives of the establishment

African Steel Mills specializes in the manufacturing and trading of steel products, including reinforcement bars, steel billets, equal angles and u-channels. The company predominantly supplies

construction companies and key distributors within Nigeria, while also exporting smaller volumes to neighbouring African countries. Its core objectives include:

- Production of high-quality steel products for the Nigerian construction industry.
- Reducing Nigeria’s dependence on imported steel through local manufacturing.
- Supporting backward integration by utilizing direct reduced iron (DRI) to reduce reliance on scrap metal.
- Continuous improvement of operational and cost effectiveness through modernization of plant technology.

2.3 Organizational Structure

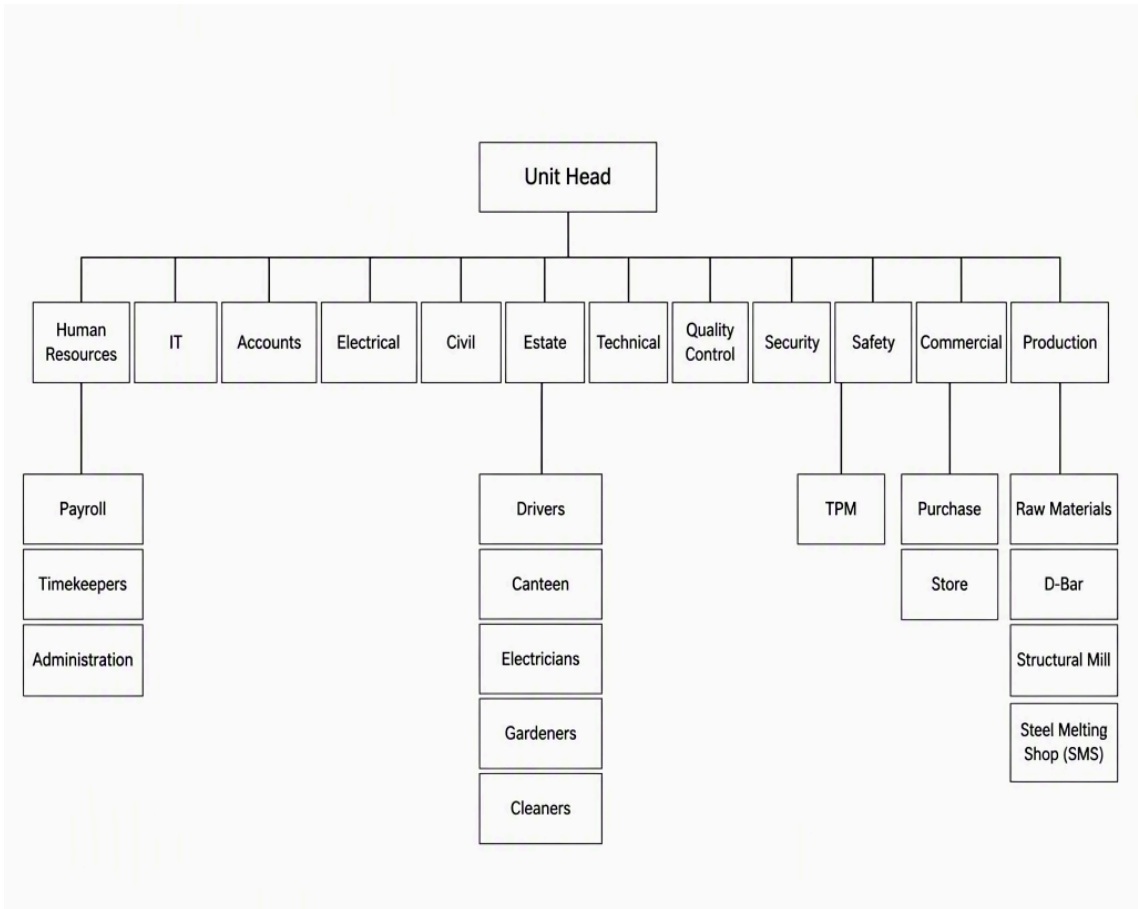


Figure 2.1: Organizational Structure of African Steel Mills Nigeria Limited

2.4 Various departments in the establishment and their functions

African Steel Mills Nigeria Ltd is organized into several departments, each with specific responsibilities that ensure smooth operations across the company. The major departments and their functions are as follows:

1. Human Resources Department: The Human Resources Department of African Steel Mills Nigeria Ltd is divided into three major units: Payroll, Timekeepers, and Administration. Each unit plays a vital role in managing staff welfare and ensuring smooth workforce operations.

- **Payroll Unit:**

This unit calculates and manages staff salaries using attendance data provided by the timekeepers. It ensures accurate payment of wages, allowances, and leave entitlements. Payroll also coordinates with the Accounts Department for proper financial record keeping.

- **Timekeepers Unit:**

The timekeepers register new employees into the system and issue clock-in cards used to record daily attendance. They monitor staff punctuality and forward attendance data to the payroll unit for salary computation.

- **Administration and Human Resource Unit:**

This unit oversees staff recruitment, interviews, suspensions, and leave approvals. It also manages external relations, including interactions with inspectors, police, and other regulatory bodies. Administration ensures discipline, compliance, and smooth communication between the company and external stakeholders.

2. Information Technology Department: The Information Technology Department is responsible for managing the company's digital infrastructure, including computer systems, networking, and software installations. It provides technical support to staff by troubleshooting devices, handling hardware upgrades, and managing user accounts. The department also oversees security systems such as CCTV cameras, biometric devices, and access control, ensuring workplace safety and staff authentication. In addition to its role within the main plant, the IT Department supervises the operations of five other branches under the ASM cluster, maintaining connectivity and ensuring uniform IT standards across all locations.

3. Account Department: The Accounts Department is responsible for managing the financial records and transactions of African Steel Mills Nigeria Ltd. It ensures proper budgeting, auditing, and documentation of all monetary activities within the company. Working closely with the payroll unit of the Human Resources Department, it receives salary data and processes payments to staff, while also maintaining accurate records for accountability. In addition, the department handles invoices, monitors expenditures, prepares financial reports, and ensures compliance with established accounting standards. By safeguarding the company's financial integrity, the Accounts Department plays a crucial role in sustaining smooth operations and supporting long-term growth.

4. Electrical Department: The Electrical Department is responsible for the installation, operation, and maintenance of all electrical systems within African Steel Mills Nigeria Ltd. It ensures uninterrupted power supply to production units, oversees the repair of electrical faults, and manages the safe use of electrical equipment across the plant. The department also supports other units by providing technical expertise in wiring, lighting, and energy distribution, thereby ensuring that machinery and facilities function efficiently and safely.

5. Civil Department: The Civil Department is responsible for all civil engineering works within African Steel Mills Nigeria Ltd. It oversees the construction, repair, and maintenance of buildings, roads, and other structural facilities in the company. The department ensures that infrastructure is safe, durable, and suitable for supporting industrial operations. By handling structural projects and maintaining the physical environment of the plant, the Civil Department plays a key role in sustaining smooth production activities and providing a safe workplace for employees.

6. Estate Department: The Estate Department is responsible for managing support services and facilities that ensure the smooth running of African Steel Mills Nigeria Ltd. It oversees a wide range of activities, from staff welfare services to the maintenance of the company's environment. The department plays a crucial role in sustaining daily operations by coordinating personnel and resources that support production and administration.

- **Drivers:** This unit manages company vehicles and transportation services, ensuring safe and efficient movement of staff, goods, and visitors.
- **Canteen:** Responsible for providing meals and refreshments to workers, supporting staff welfare and productivity.
- **Electricians:** Handle minor electrical maintenance tasks within the estate, complementing the work of the Electrical Department.
- **Gardners:** Maintain the company's green areas and landscaping, contributing to a clean and healthy environment.
- **Cleaners:** Ensure offices, workshops, and common areas are kept tidy and hygienic, supporting workplace safety and comfort.

Beyond these units, the Estate Department also supervises other auxiliary services that contribute to the overall efficiency of the company. Its broad scope of responsibilities makes it a vital support system for both staff welfare and operational effectiveness.

7. Technical Department: The Technical Department provides specialized expertise to support the production processes and machinery of African Steel Mills Nigeria Ltd. It is responsible for monitoring equipment performance, offering solutions to technical challenges, and ensuring that machines operate at optimal efficiency. By working closely with other departments, the Technical Department helps maintain smooth industrial operations, reduces downtime, and contributes to the overall productivity of the company.

- 8. Quality Control:** The Quality Control Department ensures that all steel products manufactured by African Steel Mills Nigeria Ltd meet established industrial standards. It is responsible for conducting inspections, tests, and certifications to verify product quality and durability. By monitoring production processes and identifying defects, the department safeguards the company's reputation and guarantees that only safe and reliable materials are delivered to customers.
- 9. Security Department:** The Security Department is responsible for safeguarding the assets, facilities, and personnel of African Steel Mills Nigeria Ltd. It ensures that company property is protected against theft, damage, and unauthorized access, while also maintaining order within the premises. The department manages entry and exit points, monitors activities, and works closely with other units to enforce security policies. By providing a safe and secure environment, the Security Department supports the smooth running of operations across the organization.
- 10. Safety Department:** The Safety Department is charged with ensuring a safe and healthy working environment across African Steel Mills Nigeria Ltd. It enforces compliance with safety regulations, conducts regular inspections, and promotes awareness of workplace hazards. The department also has a unit for Total Productive Maintenance (TPM) and they implement Total Productive Maintenance (TPM) practices, which focus on preventing accidents, reducing risks, and maintaining equipment reliability. Beyond TPM, the Safety Department oversees other safety initiatives and programs designed to protect employees and sustain uninterrupted industrial operations, making it a critical support system for the company's productivity and well-being.
- 11. Commercial Department:** The Commercial Department manages the procurement and storage of materials necessary for the smooth operation of African Steel Mills Nigeria Ltd. It ensures that raw materials, spare parts, and other supplies are readily available to support production and administrative activities. The department is divided into key units that handle specialized functions:

- **Purchase Unit:** Responsible for sourcing and acquiring raw materials, equipment, and services required by the company. This unit negotiates with suppliers to ensure quality and cost-effective procurement.
- **Store Unit:** Handles the receipt, storage, and issuance of materials. It maintains inventory records, monitors stock levels, and ensures that supplies are available when needed.

12. Production Department: The Production Department is the core of African Steel Mills Nigeria Ltd, responsible for transforming raw materials into finished steel products. It oversees the entire manufacturing process, ensuring efficiency, quality, and consistency in output. The department is divided into several specialized units that handle different stages of production:

- **Raw Materials Unit:** Handles the receipt, preparation, and processing of raw inputs required for steelmaking.
- **D-Bar Unit:** Focuses on producing deformed bars used in construction and other structural applications.
- **Structural Mill Unit:** Manufactures structural steel products such as beams, channels, and angles for industrial use.
- **SMS Unit:** Oversees the Steel Melting Shop, where scrap and raw materials are melted and refined into usable steel.

CHAPTER THREE: DESCRIPTION OF WORK DONE AND EXPERIENCE GAINED

3.1 Introduction

During my Students Industrial Work Experience Scheme (SIWES), I was attached to the Information Technology Department of an industrial organization where I was exposed to different areas of practical Information Technology. The training involved both routine maintenance activities and specific troubleshooting, installation, configuration and networking tasks.

The work carried out during the training covered computer hardware and software maintenance, networking, CCTV systems, printer and plotter maintenance, biometric and access-control systems, system administration, virtualization, server and storage infrastructure, Linux administration and general IT support.

A significant part of the training involved working under the supervision of experienced IT personnel. Some activities were carried out directly by me, while in other cases I assisted or observed the procedure. This provided practical experience in identifying faults, following troubleshooting procedures, handling IT equipment and understanding how different systems support the operations of an organization.

3.2 Computer Hardware Maintenance and Troubleshooting

Computer hardware maintenance was one of the recurring activities during the SIWES programme. I participated in the maintenance and handling of desktop computers and other computer equipment.

One of the activities involved opening computer systems for inspection and maintenance. I participated in cleaning computer components and subsequently reassembling the systems. Dust was removed from the internal components as part of routine preventive maintenance.

I also gained practical experience with Random Access Memory (RAM). On one occasion, I installed a RAM module into a desktop computer and reassembled the system after the installation. I also observed RAM being removed, cleaned and reseated while troubleshooting a desktop computer that was powered on but was not producing a display through HDMI.

The troubleshooting helped me understand that a display problem is not necessarily caused by the monitor or HDMI connection and that internal hardware components may also need to be investigated.

I was also exposed to storage devices such as SSDs and hard disk drives during computer maintenance. In one troubleshooting activity involving an inaccessible boot device, the system was investigated and the BIOS storage setting was eventually changed from RAID to SATA, after which the boot problem was resolved.

Other hardware-related activities included handling computer batteries, keyboards, monitors, UPS units and other peripheral equipment. I also participated in relocating computer equipment when users changed offices, ensuring that the equipment was moved and reconnected.

These activities improved my practical understanding of computer components, their physical arrangement and the importance of careful handling during installation, maintenance and troubleshooting.

3.3 Computer Software and Preventive Maintenance

Software maintenance was another recurring part of the training. I participated in routine and quarterly preventive maintenance carried out on computers in different departments.

The maintenance included clearing temporary files, clearing %temp% files and Prefetch files, and using the Disk Cleanup utility (cleanmgr) to remove unnecessary files. Group Policy was also updated during some maintenance activities.

Hardware and software maintenance were often performed together. While carrying out departmental maintenance, computer systems were physically cleaned and checked while unnecessary files and other system items were also attended to.

The preventive maintenance programme covered computers used by different departments, including Estate, Accounts, Purchase, Timekeeper, Security, Human Resources and the Learning Centre.

This experience taught me that IT support is not limited to repairing equipment after a failure. Regular maintenance can also be used to keep systems in suitable working condition and

identify possible problems before they cause greater disruption. Microsoft's documentation similarly describes Windows cleanup tools as mechanisms for removing temporary and unnecessary files.

3.4 Networking and Network Troubleshooting

Networking formed an important part of my SIWES experience. I gained exposure to Ethernet cabling, switches, routers, access points, fibre connections, SFPs, patch cords and network troubleshooting.

I learned how to terminate Ethernet cables using RJ45 connectors and a crimping tool. I arranged the conductors according to the T568B wiring arrangement, inserted them into an RJ45 connector and used the crimping tool to secure the connector. I also used a LAN cable tester to check terminated cables.

I participated in network troubleshooting activities involving switches and network connections. I used the `ping` command to test connectivity to network devices, including switches and other network destinations. At one stage, I tested the availability of more than twenty uplink switches across several company locations.

I also participated in the replacement of a malfunctioning 28-port D-Link switch with a 24-port HPE Aruba switch. The activity involved handling the switch connections and checking network connectivity after the replacement. I observed the use of a USB Type-C console connection and PuTTY to access an HPE Aruba switch for configuration and management.

I encountered fibre-optic connections and SFPs during network maintenance activities and was exposed to the use of patch cords in connecting network equipment.

In addition to workplace networking activities, I carried out practical networking exercises using Cisco Packet Tracer. These exercises included IP addressing, subnetting, DHCP, routing and connectivity testing. I also completed a two-site networking exercise involving VLANs, trunking, Router-on-a-Stick and inter-VLAN communication.

These activities helped me connect networking theory with practical network infrastructure. I gained a better understanding of how devices communicate, how connectivity faults can be isolated and how network equipment is connected and maintained.

3.5 CCTV Installation, Maintenance and Troubleshooting

I gained considerable practical exposure to CCTV systems during the training. The organization used networked CCTV systems, and my activities involved camera preparation, installation support, maintenance, troubleshooting and alignment.

I assisted in preparing CCTV cameras by unboxing and labelling them and connecting Ethernet cables between the cameras and network equipment so that the cameras could be configured by the responsible personnel.

I participated in tracing and checking cameras and identifying cameras that had gone offline. Troubleshooting involved checking possible problems with network cables, connectors, power and other parts of the camera connection.

I also participated in the repair of faulty network connections associated with CCTV cameras. On one occasion, water had entered a network cable connected to a camera. The affected connection was dried and the cable was recrimped.

In another CCTV troubleshooting activity, a damaged mounting board affected the installation of a camera. The network cable was re-crimped and a PoE splitter was used as part of the corrective work.

I also participated in replacing a camera board and using a PoE splitter during another camera repair. In another case, I helped realign a camera with a fixed and rotatable lens.

During field work at an estate location, I followed my supervisor to repair a CCTV camera whose mounting component had broken. Binding wire was used to hold the camera firmly in position, after which the camera was realigned.

I therefore gained practical experience in identifying the different possible points of failure in a CCTV installation instead of assuming that the camera itself was always the source of a problem. I learned to consider the camera, Ethernet connection, connector, PoE arrangement, switch and physical mounting when troubleshooting a networked camera.

3.6 Printer and Plotter Maintenance

Printer and plotter maintenance formed part of my regular IT support activities.

I carried out routine maintenance on printers and the Canon plotter at the Learning Centre. I also performed printer checks and attended to printers in different departments.

I gained experience installing printer software and testing printers after installation. In one case, I installed the required software for two printers and carried out test prints to confirm that they were functioning.

I also participated in the installation of an HP LaserJet Pro M479 printer on a user's computer through a wired network connection. This helped me understand the difference between directly connected printers and network-connected printers.

I also encountered USB-connected printers and learned that a USB-connected printer does not use an IP address in the same way as a network-connected printer.

The maintenance and installation activities gave me practical experience in supporting office printing equipment and troubleshooting peripheral devices.

3.7 Biometric and Access-Control Systems

I was exposed to biometric and electronic access-control systems used within the organization.

Some biometric devices were used for employee clock-in and clock-out, while others were associated with physical access control. I worked around biometric devices that experienced network connectivity problems and participated in activities involving their network connections.

I also encountered facial-recognition systems used to control access to rooms and doors. The system authenticates authorized individuals before access is granted.

Another access-control system I encountered was a flap barrier. I participated in checking the flap barrier and observed the testing of its fingerprint function. I also participated in work involving an exit button for an access-control system in the server room.

I learned that biometric systems can form part of a wider IT infrastructure because their operation may depend on network connectivity, authentication systems and physical access-control equipment.

3.8 System Administration and Virtualization

I gained exposure to Windows system administration and virtualization during the training.

I used and observed Windows administrative utilities such as Command Prompt, PowerShell, Group Policy and system configuration tools. I learned commands for retrieving system information, including the BIOS serial number of a computer.

VirtualBox was also used to create virtual machines for practical exercises. I used virtual machines to practise system administration and networking tasks without requiring separate physical computers for every system.

Virtualization was particularly useful during my Active Directory exercises, where a virtual machine was used to create and configure an Active Directory environment.

3.9 Active Directory and Windows Domain Administration

I carried out practical exercises involving Active Directory in a virtualized environment.

I created and configured an Active Directory environment using a virtual machine and worked with a Domain Controller. I also learned about users, computers, organizational units and security groups.

I configured the virtual networking required for another virtual machine to communicate with the Active Directory environment. The second virtual machine was hosted on another physical computer, providing practical exposure to communication between virtual machines across separate physical systems.

I successfully joined a user/computer to the Active Directory domain. During this process I encountered the importance of correct network and DNS configuration in enabling a computer to locate and communicate with a Domain Controller.

Active Directory provides centralized management of network objects such as users and computers, while Group Policy provides centralized management of user and computer settings.

The exercise gave me practical exposure to concepts that I had previously encountered mainly in theory, particularly domain controllers, domain joining, DNS, user management and centralized administration.

3.10 Linux User Management and File Permissions

I also carried out a practical exercise using Ubuntu Linux.

I created a file named `confidential` and configured different users with different levels of access. One user was given full permissions, another was given permission to read and modify the file, while another user was given read-only access.

This exercise helped me understand the practical use of Linux file permissions and user authorization. It also demonstrated how access to information can be restricted according to the responsibilities assigned to different users.

3.11 Server, Storage and Infrastructure Support

I was exposed to activities involving the server room and supporting infrastructure.

I recorded the status of the server room and observed the importance of maintaining the environment and equipment used to provide IT services.

I also encountered Synology as part of the organization's storage infrastructure. I gained exposure to its role in the organization's IT environment and observed how backup power supported critical infrastructure during a power outage.

Other infrastructure-related systems I encountered included Avaya communication equipment and network infrastructure connecting different parts of the organization.

These activities helped me understand that enterprise IT involves more than individual computers. Servers, storage systems, network equipment, communication systems and power-support equipment work together to keep organizational services available.

3.12 IT Support, Documentation and Equipment Management

General IT support and documentation formed part of my daily experience.

I assisted with setting up users' workstations, relocating office equipment and resolving user-related technical problems. I also updated and handled reports as part of workplace documentation.

I participated in IT equipment inventory activities, including identifying equipment and recording relevant information such as serial numbers. I learned methods of retrieving computer identification information through Windows command-line and PowerShell tools.

I also participated in retrieving equipment assigned to a retired staff member, including a desktop computer, scanner, UPS and Avaya telephone. This exposed me to the importance of accounting for organizational IT equipment.

I also handled routine reports and documentation during the training. The compilation records report preparation as a genuine activity but also distinguishes it from technical work; therefore, I have treated documentation as supporting IT work rather than presenting report preparation itself as a major technical achievement.

3.13 Experience Gained

The SIWES programme gave me practical experience in several areas of Information Technology.

I developed greater familiarity with computer hardware through RAM installation, hardware cleaning, system reassembly, storage-device handling and troubleshooting.

I gained practical networking experience through Ethernet cable termination, cable testing, switch maintenance, connectivity testing, fibre-related work and network simulation. The networking practicals also helped me understand subnetting, DHCP, VLANs, routing and inter-network communication.

The CCTV activities improved my understanding of networked surveillance systems and taught me to approach faults systematically by considering both the physical and network components of a system.

I also gained experience with printers, plotters, biometric devices, access-control systems and enterprise communication equipment.

The system-administration exercises introduced me to Active Directory, Windows domain environments, Group Policy, virtualization and Linux file permissions.

An important experience gained throughout the training was the development of a more systematic approach to troubleshooting. Instead of immediately assuming that a device itself was faulty, I learned to consider the components and connections that support it and to test possible causes progressively.

The training also demonstrated the importance of continuous learning in IT. Many of the activities required consultation with experienced personnel, technical information and practical investigation. This reinforced the understanding that IT professionals must continue learning as they encounter unfamiliar systems and problems.

3.14 Summary

The work carried out during the SIWES programme covered computer hardware and software maintenance, networking, CCTV systems, printer and plotter maintenance, biometric and access-control systems, system administration, virtualization, Active Directory, Linux administration, server and storage infrastructure, documentation and general IT support.

The training provided practical exposure to both routine maintenance and problem-solving activities. More importantly, it enabled me to connect theoretical concepts from my Information Technology studies with actual equipment and systems used within an industrial organization.

The experience strengthened my practical understanding of IT infrastructure and improved my ability to handle equipment, test connectivity, investigate faults, perform routine maintenance and work under the supervision of experienced IT personnel.

CHAPTER FOUR: EQUIPMENT AND TOOLS USED, THEIR FUNCTIONS AND DETAILED DESCRIPTION OF USAGE

4.1 Introduction

During my SIWES programme, I was exposed to a variety of hardware, networking equipment, security systems, communication devices, software tools, and other IT equipment used in the daily operations of the organization. These tools were used for network connectivity, computer maintenance, surveillance, access control, communication, printing, system administration, troubleshooting, and general technical support.

4.2 Networking Equipment and Tools

Table 4.1 — Summary of Equipment and Tools

Equipment/Tool	Category
Network Switch	Networking Tools
Router	
SFP	
LAN Tester	
Patch Cord	
Crimping Tool	
Access Point	

Ethernet Cable and RJ45 Connector	
USB Type-C Console Connection	
PuTTY	
CCTV Camera	CCTV Systems
NVR	
Hard Drive	
RAM	Computer Hardware
SSD/HDD	
UPS	
Plotter	
Printer	
Avaya	Telephony

Cisco Packet Tracer	Software
Oracle Virtual Box	
VNC	
Genian	
Synology	
Windows 10 - 11	Operating System
Linux	
Windows Server	

4.2.1 Network Switch

A network switch is a networking device used to connect multiple devices within a Local Area Network (LAN) and facilitate communication between them. It receives data from connected devices and forwards it through the appropriate network port to the intended destination. Switches are commonly used to connect computers, servers, printers, access points, CCTV systems, and other network devices.

During my SIWES training, I encountered network switches as an important part of the organization's network infrastructure. I participated in activities involving network connectivity testing and troubleshooting. I used the **ping** command to check the reachability of switches, servers, and other network devices and was involved in checking the connectivity of numerous uplink switches across different locations.

I also participated in the replacement of a faulty 28-port D-Link switch with a 24-port HPE Aruba switch. This exposed me to practical switch replacement and the importance of properly reconnecting network cables and confirming network connectivity after maintenance. I also observed the use of an HPE Aruba switch through a USB Type-C console connection and PuTTY for accessing the switch interface.

Examples of switches encountered during the training included D-Link and HPE Aruba switches.



Figure 4.1: Network Switch

4.2.2 SFP

An SFP (Small Form-factor Pluggable) is a removable network transceiver used to provide a network interface for devices such as switches. Depending on the type of SFP and the connected medium, it can be used to transmit and receive data through fibre-optic or other compatible network connections.

During my SIWES training, I encountered SFPs while working with the organization's network infrastructure. I observed their use in network connections and participated in activities involving the replacement of an SFP during network maintenance. I also worked with fibre connections and learned how SFPs interface with network equipment to provide communication between different points on the network.

4.2.3 Network Patch Cords

Network patch cords are short network cables used to connect network equipment such as switches, patch panels, computers, and other devices. They provide a convenient physical connection between network components.

During my training, I worked with network patch cords when connecting and reconnecting network equipment during maintenance and troubleshooting. I encountered fibre patch cords, including LC-to-LC and LC-to-SC connections, and learned how the correct patch-cord type and connection are important for establishing communication between network devices. Patch cords were also handled during switch replacement and other network maintenance activities.



Figure 4.2: Fibre-Optic Patch Cord



Figure 4.3: Network Patch Cord

4.2.4 RJ45 Crimping Tool

An RJ45 crimping tool is a hand tool used to attach RJ45 connectors to Ethernet cables. It secures the connector onto the cable and establishes the physical connection between the cable conductors and the connector contacts.

During my SIWES training, I used the crimping tool when terminating Ethernet and Cat6 cables. I arranged the cable conductors according to the required wiring arrangement, inserted them into an RJ45 connector, and used the crimping tool to secure the connector. I applied this skill during network, CCTV, and biometric-system maintenance. For example, I re-crimped a damaged Ethernet cable connected to a biometric device and tested the connection afterward.



Figure 4.5: RJ45 Connector

4.2.5 Ethernet Cable and RJ45 Connector

Ethernet cables are used to establish wired network connections between devices, while RJ45 connectors provide the physical termination used to connect Ethernet cables to network ports.

During my training, I worked with Ethernet and Cat6 cables in networking, CCTV, and biometric-system activities. I learned how to prepare and terminate Ethernet cables by arranging the internal wires correctly and attaching RJ45 connectors using a crimping tool. I also used a LAN cable tester to check whether terminated cables were properly connected.

I applied these skills when working on CCTV cameras and biometric devices. For example, I crimped an Ethernet cable for a biometric device that had been physically damaged, allowing the device's network connection to be restored. I also worked with network cables during CCTV troubleshooting and other network maintenance activities.



4.2.6 LAN Cable Tester

A LAN cable tester is a testing tool used to determine whether the individual conductors in an Ethernet cable are correctly connected and whether the cable has continuity.

During my SIWES training, I used a LAN cable tester when working with Ethernet cables and RJ45 connectors. After terminating cables, the tester was used to verify that the conductors were properly connected. This was particularly useful during network, CCTV, and biometric-device maintenance because a faulty cable or incorrect termination can prevent a device from communicating with the network.

4.2.7 Router

A router is a networking device used to connect different networks and direct data packets between them.

During my SIWES training, I worked with routers while learning and practising network configuration and troubleshooting. I observed how routers were used to facilitate communication between different network segments and provide connectivity between devices on separate networks. I also used routers during networking practical exercises involving IP addressing, DHCP and inter-network communication.

4.2.8 Access Point

An access point (AP) is a networking device that provides wireless connectivity to devices and connects them to a wired network.

During my training, I worked with access points when setting up and troubleshooting wireless network connectivity. I observed the configuration of wireless network parameters and the connection of wireless devices to the company's network. In one Cisco Packet Tracer exercise, Site A contained four access-point clients. This helped me understand how wireless clients can be incorporated into a network. Access points were also relevant when diagnosing connectivity problems involving wireless users.

4.2.9 USB Type-C Console Connection

A USB Type-C console connection provides a direct connection between a computer and the console interface of a network device for configuration and management.

I observed an HPE Aruba switch being accessed through a USB Type-C console connection. The connection was used with PuTTY to access the switch's console interface. This introduced me to the use of direct console access when configuring or troubleshooting network equipment.

4.2.10 PuTTY

PuTTY is a terminal emulator and remote-access application commonly used to establish console, SSH and other communication sessions with network devices.

During my training, I observed the use of PuTTY together with the USB Type-C console connection to access the command-line interface of the HPE Aruba 6000 switch. Through the console session, they were able to observe and work with the switch's configuration and networking commands.

4.3 CCTV System

A Closed-Circuit Television (CCTV) system is a surveillance system used to monitor and record activities within designated areas. A typical CCTV system consists of cameras, network connections, power components, and a Network Video Recorder (NVR) for managing and recording video feeds.

During my SIWES training, I was exposed to Hikvision CCTV systems and participated in different CCTV installation, maintenance, and troubleshooting activities. I assisted with tracing and mapping cameras to their corresponding NVR channels and learned how cameras communicate through the network infrastructure.

I also participated in troubleshooting cameras that had gone offline. During these activities, possible causes such as damaged or poorly crimped Ethernet cables, PoE problems, damaged camera connections, and physical mounting problems were investigated. I worked with Ethernet cables and PoE components and participated in reconnecting and testing cameras.

I also assisted during camera alignment activities. In one case, a camera was realigned to achieve the appropriate viewing position. In another field activity, I followed my supervisor to repair an estate CCTV camera whose mounting component had broken. Binding wire was used to hold the camera firmly in position, after which the camera was realigned.

The CCTV equipment encountered included different types of IP cameras, such as bullet, dome, and mini-PTZ cameras, as well as NVRs and related network and power components.



Figure 4.6: CCTV Camera and Network Video Recorder (NVR) Equipment

4.4 Computer Hardware

Computer hardware refers to the physical components that make up a computer system. During my training, I worked with and observed several internal computer components, including RAM, SSDs, HDDs, batteries, CPUs, and other components.

RAM (Random Access Memory) provides temporary working memory for a computer and allows the operating system and applications to access data needed during operation. During hardware maintenance activities, I participated in installing RAM and learned how the component is properly fitted into the motherboard.

An SSD (Solid-State Drive) is a storage device used to permanently store the operating system, applications, and user data. I encountered SSDs during computer hardware maintenance and installation activities. I participated in the installation and replacement of storage drives and helped reassemble computers after hardware work. The systems were subsequently checked to ensure that they could boot and operate correctly.

I also participated in computer maintenance involving the cleaning of CPUs and other internal components. Dust was removed from systems using appropriate cleaning procedures before the computers were reassembled. I also encountered computer batteries and participated in hardware replacement activities.

4.5 Uninterruptible Power Supply (UPS)

A UPS is a power-protection device that provides temporary backup power to connected equipment when the main electricity supply fails. It also helps protect equipment against certain power-related problems.

During my SIWES training, I encountered UPS units connected to computers and other IT equipment. I participated in office equipment relocation where UPS units were moved and reconnected alongside computers and other equipment. I also encountered UPS units during the retrieval of equipment belonging to a retired staff member.

The importance of backup power was also observed in the server room. During a prolonged power outage, the backup power system supported critical IT services, including network services, Navision, and Synology, helping to keep them operational.

4.6 Printer

A printer is an output device used to produce physical copies of digital documents. Printers are commonly used in offices to produce reports, forms, correspondence, and other documents.

During my SIWES training, I participated in the maintenance, installation, configuration, and troubleshooting of printers. I carried out routine maintenance on printers at the Learning Centre and other departments. I also connected a laptop to a printer using a USB cable, installed the required printer software, configured the printer, and printed a test page to confirm that the installation was successful.

I also participated in the installation of an HP LaserJet Pro M479 on a user's computer through a wired network connection. This gave me practical experience with both direct and network-based printer setup.

4.7 Plotter

A plotter is a specialized output device used to produce large-format prints, particularly for technical drawings, plans, diagrams, and other documents that require a larger paper size.

During my SIWES training, I regularly carried out maintenance on the Canon plotter at the Learning Centre. The maintenance formed part of the routine care of the equipment and was intended to keep it functional and available when required. I also learned about proper startup and handling procedures for the equipment.



Figure 4.7: Canon Plotter

4.8 Avaya Telephony System

Avaya is an enterprise communication and telephony system used to provide business voice communication and related communication services.

During my SIWES training, I encountered Avaya equipment during system maintenance and office relocation activities. On one occasion, an Avaya device displayed an administrator password interface. The appropriate administrator credentials were entered, after which the device was restarted and a software update was observed. I also had to log back into the system following a server update.

Avaya equipment was also included among the equipment moved and reconnected during an office relocation exercise. This gave me practical exposure to enterprise telecommunication equipment and its integration with the organization's IT infrastructure.



Figure 4.8: Avaya Telephone

4.9 Biometric and Access-Control Systems

Biometric systems use unique biological characteristics, such as fingerprints or facial features, to identify or authenticate individuals. They can be used for employee attendance, access control, and movement management. Access-control systems may also be connected to physical devices such as doors and flap barriers.

During my SIWES training, I encountered different biometric systems used for different purposes within the organization. Some biometric devices were used for **clocking in and out**, allowing employees' attendance and working hours to be recorded. I assisted with troubleshooting such devices, including resolving network-related problems and re-crimping Ethernet cables when necessary.

I also encountered **facial-authentication systems used for controlling access to doors and rooms**. These systems were used to authenticate authorized personnel before granting access. I worked around the network and physical connections associated with these access-control devices and gained practical exposure to the interaction between biometric authentication and physical access-control equipment.

Another access-control system I encountered involved a **flap barrier**. The biometric authentication system communicates with the access-control mechanism, and when authorization is successful, the barrier permits the individual to pass. I also participated in the installation of an exit button for a biometric device located in the server room.

These activities helped me understand that IT support can extend beyond computers and networks to include physical security, biometric authentication, access control, and the communication between different electronic systems.



Figure 4.9: Flap Barrier Access-Control System

4.10 Cisco Packet Tracer

Cisco Packet Tracer is a network simulation tool used to create and test virtual network environments without requiring the physical networking equipment.

During my SIWES training, I used Packet Tracer for practical networking exercises. I created network topologies involving computers, switches, routers, servers, and access points. I practiced subnetting, DHCP configuration, and connectivity testing. I also worked on a two-site network involving VLANs, Router-on-a-Stick, trunking, inter-VLAN communication, DHCP, and communication between sites.

The simulations allowed me to practice network configuration and troubleshooting concepts before applying similar principles to real networking environments.

4.11 Ping Command

The **ping** command is a network diagnostic tool used to test whether a device or network destination is reachable and responding to network requests.

During my SIWES training, I used **ping** as part of network troubleshooting and connectivity testing. I used it to check communication with switches, servers, and other network devices. I also tested external destinations such as **8.8.8.8** and network service providers. These tests helped determine whether connectivity existed between the source computer and the destination and assisted in identifying where network problems could be occurring.

4.12 NSLOOKUP Command

nslookup is a network diagnostic command used to query the Domain Name System (DNS) and determine how domain names are resolved to IP addresses.

I encountered **nslookup** while learning and troubleshooting Active Directory and DNS. During one practical exercise, the IP address of the Domain Controller could be reached through ping, but the domain name could not be resolved correctly using **nslookup**. This helped me understand the importance of correct DNS configuration in an Active Directory environment.

4.13 Windows System Maintenance Tools

Windows system maintenance tools and utilities are used to remove unnecessary files, maintain system performance, and manage system settings.

During quarterly preventive maintenance, I used or participated in activities involving `cleanmgr`, `%temp%`, and Prefetch files. These were used to remove unnecessary temporary data and perform routine system maintenance. I also participated in updating Group Policy and carrying out general system checks.

These activities were performed on computers across different departments as part of the organization's preventive maintenance programme. Hardware maintenance, including cleaning dust from computer components, was also carried out alongside software maintenance.

4.14 Active Directory

Active Directory is a Microsoft directory service used to centrally manage users, computers, groups, authentication, and other resources within a Windows domain environment.

During my SIWES training, I created and configured an Active Directory environment using a virtual machine. I worked with a Domain Controller and learned about organizational units, users, computers, servers, and security groups.

4.15 Virtual Machine and VirtualBox

A virtual machine is a software-based computer environment that allows an operating system and applications to run on a physical computer as though they were operating on a separate computer. VirtualBox is virtualization software used to create and manage virtual machines.

During my training, I used virtual machines as part of my Active Directory practical exercises. I created a Domain Controller in a virtual machine and configured another virtual machine to communicate with it. I also worked with virtual machines running on separate physical computers, which provided practical exposure to virtual networking and domain connectivity.

Virtualization allowed me to practice system administration and networking concepts in a controlled environment without requiring multiple physical servers or computers.

4.16 Ubuntu Linux

Ubuntu is a Linux-based operating system used for computing, system administration, software development, networking, and other technical tasks.

During my SIWES training, I used Ubuntu to practice Linux user management and file permissions. I created a file named **confidential** and configured different levels of access for three users. One user was given administrator-level permissions, another was given permission to read and modify the file, while the third user had read-only access.

This exercise helped me understand Linux file permissions, access control, user authorization, and the principle of giving users only the level of access required for their role.

4.17 Synology

Synology is a storage and network infrastructure platform commonly used for centralized data storage, backup, and related network services.

During my SIWES training, I was exposed to Synology as part of the organization's server and storage infrastructure. I encountered it during server-room activities and learned about its role in supporting organizational data and IT services.

During a power outage, the server-room backup power system helped keep Synology and other critical services operational. This demonstrated the relationship between storage infrastructure, network availability, and reliable power supply.

4.18 Genian

Genian was an endpoint-related application/system encountered during my SIWES training. It was relevant to the operation and network access of users' computers within the organization.

I encountered several network-related issues associated with users being logged out of or having Genian disabled. During user support activities, I assisted in resolving these problems so that affected computers could regain normal network access. This showed me that enterprise endpoint applications can be integrated with an organization's network access and security arrangements and that problems with such applications can affect users' ability to connect to network resources.

4.23 VNC

VNC (Virtual Network Computing) is a technology that allows a computer to be accessed and controlled remotely over a network.

During my SIWES training, I was exposed to remote computer access as part of IT support activities. Remote access allowed IT personnel to work on or provide support for computers without necessarily being physically present at the user's workstation. This provided practical understanding of how remote administration can be used for troubleshooting, configuration, and technical support.

4.24 Summary

The equipment and tools encountered during the SIWES programme provided practical exposure to several areas of Information Technology. Networking equipment such as switches, fibre components, Ethernet cables, and testing tools introduced me to enterprise network infrastructure and troubleshooting. CCTV and biometric systems exposed me to surveillance and physical access-control technologies, while computer hardware and maintenance tools provided practical experience in servicing computer systems.

I also gained experience with software and system-administration tools such as Cisco Packet Tracer, Active Directory, VirtualBox, Ubuntu, Windows maintenance utilities, and remote-support technologies. In addition, printers, plotters, Avaya systems, Synology, and other enterprise technologies demonstrated how different IT systems work together to support the daily operations of an organization.

CHAPTER FIVE: SUMMARY, CONCLUSION AND RECOMMENDATIONS

5.1 Summary of Attachment Activities

The Students Industrial Work Experience Scheme (SIWES) provided me with practical exposure to Information Technology operations within an industrial environment. I carried out my industrial attachment in the Information Technology Department of African Steel Mills Nigeria Limited, where I worked under the supervision and guidance of experienced IT personnel.

During the training, I participated in computer hardware and software maintenance activities. These included cleaning computer systems, installing RAM, handling storage devices, troubleshooting hardware problems, carrying out preventive maintenance, and performing routine software maintenance such as clearing temporary files and using Windows maintenance utilities. I also participated in setting up and relocating users' computer systems and resolving various technical problems.

Networking was another major area of the training. I gained practical experience with Ethernet cables, RJ45 connectors, crimping tools, LAN cable testers, network switches, routers, access points, SFPs and network patch cords. I participated in cable termination, cable testing, connectivity testing, switch replacement and other network troubleshooting activities. I also used commands such as `ping` and `nslookup` during connectivity and DNS-related exercises. In addition, I carried out networking practicals with Cisco Packet Tracer involving IP addressing, subnetting, DHCP, VLANs, routing and inter-VLAN communication.

I also gained practical exposure to CCTV systems. I participated in the installation support, maintenance and troubleshooting of networked CCTV cameras. The activities included checking offline cameras, working with Ethernet connections and PoE components, repairing damaged cable connections, assisting with camera replacement and alignment, and supporting physical mounting repairs. These activities helped me understand the relationship between CCTV equipment and the network infrastructure supporting it.

The training also covered printers and plotters. I participated in routine printer maintenance, printer installation, software installation and test printing. I also worked with the Canon plotter at the Learning Centre as part of its routine maintenance.

Another area of exposure was biometric and access-control technology. I encountered systems used for employee attendance, facial authentication and physical access control. I participated in troubleshooting network-related problems with biometric devices and was exposed to systems such as the flap barrier and access-control devices.

I was further exposed to system administration and virtualization. I used VirtualBox for practical exercises involving Active Directory and virtual machines. I worked with a Domain Controller, users, computers, organizational units and security groups, and gained experience

with domain joining and the importance of network and DNS configuration. I also used Ubuntu Linux to practise user management and file permissions.

Other areas of exposure included server-room infrastructure, Synology storage, Avaya telephony, VNC remote support, Genian, equipment inventory and general IT documentation. These activities gave me a broader understanding of how different IT systems operate together within an organization.

Overall, the SIWES programme helped me relate the theoretical knowledge acquired in school to practical situations. It improved my understanding of IT equipment, networking, troubleshooting, preventive maintenance, system administration and technical support. It also exposed me to the importance of working under supervision, following proper procedures, documenting activities and continuously learning when dealing with unfamiliar technologies.

5.2 Problems Encountered During the Programme

During the SIWES programme, I encountered several challenges while carrying out practical activities. Some of these challenges were technical, while others were related to working with unfamiliar equipment and systems.

One of the technical challenges encountered was troubleshooting faulty computer systems. Problems such as display failure and inaccessible boot devices required investigation of different hardware and system settings before the actual cause could be identified. These situations showed that a fault may not always originate from the component that initially appears to be responsible.

Network-related problems were also encountered during the training. Some devices experienced connectivity problems due to damaged cables, connectors, switches or other network components. CCTV cameras and biometric devices could also become unavailable when their network connections or supporting equipment developed faults. Troubleshooting such problems sometimes required checking several components before the cause could be identified.

Another challenge was working with unfamiliar enterprise systems and equipment. Systems such as Active Directory, network switches, Synology, Avaya, Genian and access-control equipment required additional explanation and practical guidance before their operation could

be properly understood. Some activities therefore required close supervision and consultation with experienced IT personnel.

Power interruptions were another challenge within the industrial environment. The continued operation of important IT infrastructure during power outages depended on backup power systems. This demonstrated the importance of reliable power support for maintaining the availability of critical IT services.

There were also occasions when repairs required replacement parts or physical corrective measures. For example, damaged cables, connectors and CCTV mounting components had to be repaired or replaced before the affected equipment could function properly.

Despite these challenges, they contributed to the learning experience. They encouraged careful observation, systematic troubleshooting, patience and consultation with experienced personnel rather than attempting to solve unfamiliar problems without adequate knowledge.

5.3 Suggestions for Improvement of the Scheme

Based on the experience gained during the SIWES programme, the following suggestions may help improve the effectiveness of the scheme:

1. **Better coordination between institutions and host organizations:** Universities and ITF should maintain regular communication with participating organizations to ensure that students are properly supervised and exposed to activities relevant to their courses of study.
2. **More practical preparation before industrial attachment:** Students should receive adequate practical orientation before beginning their attachment. Basic training in networking, hardware maintenance, operating systems and troubleshooting would make it easier for students to participate effectively in industrial activities.
3. **Greater student involvement in practical tasks:** Where possible, students should be allowed to participate directly in suitable technical activities under supervision rather than only observing the work. This would increase their practical confidence and understanding.
4. **Improved monitoring by institutions:** University supervisors should maintain regular contact with students and their host organizations throughout the attachment

period. This would help ensure that students are gaining relevant experience and allow problems to be identified early.

5. **Access to a wider range of IT technologies:** Students should be exposed to different categories of IT equipment and software where available. Exposure to networking equipment, security systems, server technologies, virtualization and system administration can provide a broader understanding of professional IT practice.
6. **Proper documentation of students' activities:** Students should be encouraged to maintain accurate daily records of the work they perform and the knowledge they gain. This would make the preparation of SIWES reports easier and provide a useful record of their industrial experience.
7. **Adequate duration and structured training:** The attachment should provide sufficient time for students to move beyond observation and gradually take part in practical tasks under supervision. A structured progression from basic activities to more advanced tasks would make the training more effective.

5.4 Conclusion

The Students Industrial Work Experience Scheme provided valuable practical exposure to the application of Information Technology in an industrial environment. The training covered computer maintenance, networking, CCTV systems, biometric and access-control systems, printing equipment, system administration, virtualization, Linux, server and storage infrastructure, and general technical support.

The experience helped me develop a better understanding of how IT systems are installed, maintained, connected and supported in an organization. It also improved my ability to approach technical problems systematically and demonstrated the importance of continuous learning and practical experience in Information Technology.

The programme therefore served as an important link between academic knowledge and industrial practice. The knowledge and practical experience gained during the attachment will provide a useful foundation for further development in the field of Information Technology.

